

Number Sequences

Worksheet 3 (Calculator)

1. S14/qp21/q5

- a. Here is part of a number grid. A square can be placed anywhere on the grid outlining four numbers. The numbers in opposite corners of the square are multiplied together and the difference between the products is found.

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

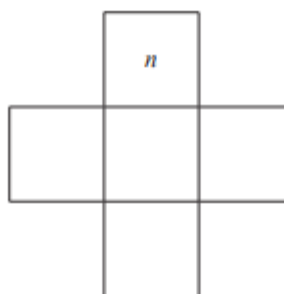
$$9 \times 14 - 8 \times 15 = 126 - 120 = 6$$

- (i) The grid is continued downwards. If n represents the number of the top left of the square, complete this square with expressions for the other numbers. [1]
- (ii) Use your answer to part (a)(i) to prove that the difference between the products of the opposite corners is always 6. [2]

- b. Here is part of a different number grid. A cross can be placed anywhere on the grid outlining five numbers. The numbers in the cross are added together.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36				
41	42								

- (i) Find and simplify an expression, in terms of n , for the sum of the numbers in the cross below. [2]



- (ii) The sum of the numbers in the cross below is 330. Complete the cross with the correct numbers. [2]

2. S14/qp22/q6

- a. The first five terms of a sequence are 17, 11, 5, -1, -7.

Find, in terms of n , an expression for the n th term of this sequence. [2]

- b. The n th term, S_n , of a different sequence is found using the formula $S_n = n^2 + 3n$.

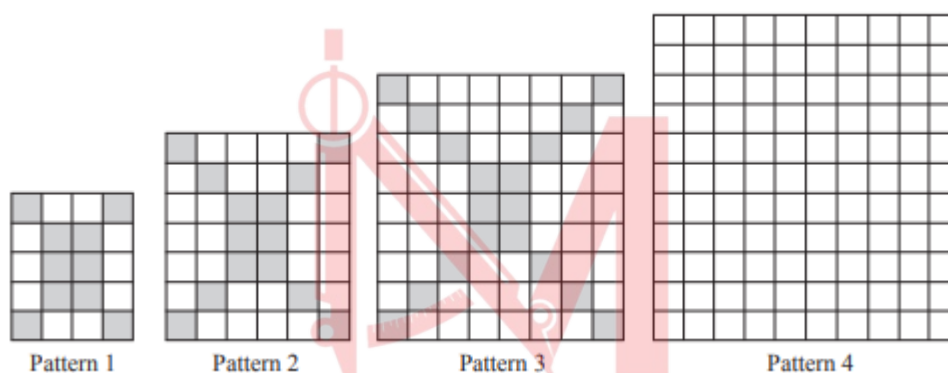
i. Work out the first four terms of this sequence [2]

ii. The n th term, T_n , of another sequence is found using the formula $T_n = 5n - 12$

. There are two values of n for which $\frac{S_n}{T_n} = 6$. Form and solve an equation in n to

find these two values. [4]

3. S17/qp21/q8



- a. Complete the diagram for pattern 4. [1]

The table below shows some of the information for the number of tiles in pattern n .

Pattern n	1	2	3	4	5
Number of grey tiles	10	14	18		
Number of white tiles	10	28	54		
Total number of tiles	20	42	72	110	156

- b. Complete the table. [2]

c. Find an expression, in terms of n , for the number of grey tiles in pattern n . [2]

d. Pattern x has 110 grey tiles. Find x . [1]

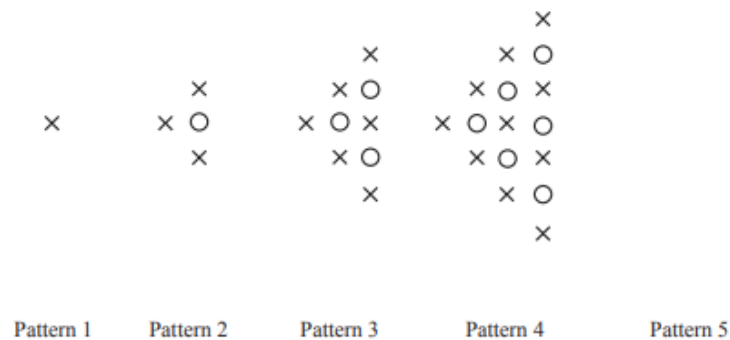
e. By considering the number of tiles along the outer edges of each pattern, show that the total number of tiles in pattern n is $4n^2 + 10n + 6$. [2]

f. Hence find an expression, in terms of n , for the number of white tiles in pattern n . [1]

g. In pattern p , the total number of tiles is equal to 8 times the number of grey tiles. Find p .

[3]

4. S17/qp22/q7



The diagrams show patterns made from crosses () and circles ().

- a. Draw pattern 5 above. [1]

The table shows the number of crosses and circles in each pattern.

Pattern number (n)	1	2	3	4	5	6
Number of crosses	1	3	6	10		
Number of circles	0	1	3	6		
Total number of crosses and circles	1	4	9	16	25	36

- b. Complete the table. [2]

- c. Find an expression, in terms of n , for the total number of crosses and circles in pattern n . [1]

- d. An expression, in terms of n , for the number of crosses in pattern n is $\frac{1}{2}n^2 + \frac{1}{2}n$. How many crosses are there in pattern 30? [1]

- e. Show that the number of circles in pattern n is $\frac{1}{2}n^2 - \frac{1}{2}n$. [1]

- f. The number of crosses in pattern m is equal to 5m. Find m . [3]

5. S18/qp21/q5

- a. 1,7,13,19,25

- i. Find an expression, in terms of n , for the n th term of this sequence. [2]

- ii. Explain why 251 is not a term in this sequence. [1]

- b. Here is another sequence. 5, 8, 13, 20, 29, ... The p^{th} term of this sequence is $p^2 + 4$. Write down an expression, in terms of p , for the p^{th} term of these sequences.

- i. -2, 1, 6, 13, 22, [1]

- ii. 7, 12, 19, 28, 39, [1]

6. W18/qp21/q8

The diagram shows part of a number grid.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

A vertical rectangle enclosing three numbers, as shown, can be placed anywhere on the grid.

The grid is continued downwards.

- a. If n represents the number in the top of the rectangle, complete the rectangle with expressions, in terms of n , for the other two numbers. [1]



- b. Omar multiplies the top number in the rectangle by the bottom number. He then squares the middle number in the rectangle. He finds the difference between these two results. Using your answers to part (a), show that this difference is always 25. [2]
- c. Lena places a rectangle on the grid. She adds the three numbers in her rectangle. The sum of the three numbers is 174. Find the largest number in Lena's rectangle. [3]

Answer Key

1. S14/qp21/q5

5	(a) (i)	$n + 6, n + 7$	1	
	(ii)	$(n + 1)(n + 6) - n(n + 7)$ $= n^2 + 7n + 6 - n^2 - 7n = 6$	2	M1 for $(n + 1)(n + 6) - n(n + 7)$ or reversed Or B1 for $n^2 + 7n + 6$
	(b) (i)	$5n + 50$ or $5(n + 10)$	2	M1 for $[n], n + 9, n + 10, n + 11, n + 20$ seen
	(ii)	56, 65, 66, 67, 76 completed in cross	2	M1 for $n = 56$ Or for 66 in centre of cross

2. S14/qp22/q6

6	(a)	$23 - 6n$ cao	2	B1 for $-6n$ soi
	(b) (i)	4, 10, 18, 28	2	B1 for 3 correct terms seen
	(ii)	3 and 24	4	M1 for $\frac{n^2 + 3n}{5n - 12} = 6$ or better M1 for $n^2 - 27n + 72 = 0$ B1 for either 3 or 24

3. S17/qp21/q8

8(a)	Correct diagram	1	
8(b)	$\begin{array}{cc} 22 & 26 \\ 88 & 130 \end{array}$	2	B1 for 2 or 3 correct
8(c)	$4n + 6$ oe isw	2	B1 for $4n \pm k$
8(d)	26	1	
8(e)	$(2n + 3)(2n + 2)$ leading to $4n^2 + 10n + 6$ with no errors	2	B1 for either $(2n + 3)$ or $(2n + 2)$ used After 0, SC1 for $4n^2 + 10n + 6$ shown using alternative method
8(f)	$4n^2 + 6n$ oe	1	
8(g)	7 cao	3	M1 for $4p^2 + 10p + 6 = 8 \times \text{their } (4p + 6)$ A1 for $4p^2 - 22p - 42 [= 0]$ oe or B2 for $[p = 7]$ total 272 grey 272 or B1 for $[p = 6]$ total 240 grey 240

4. S17/qp22/q7

7(a)	Correct pattern drawn	1	
7(b)	15 21 10 15	2	B1 for 2 or 3 correct
7(c)	n^2 oe final answer	1	e.g. $(\frac{1}{2}n^2 + \frac{1}{2}n) + (\frac{1}{2}n^2 - \frac{1}{2}n)$
7(d)	465	1	
7(e)	$n^2 - (\frac{1}{2}n^2 + \frac{1}{2}n)$ or $(\frac{1}{2}(n-1)^2 + \frac{1}{2}(n-1))$ or $(\frac{1}{2}n^2 + \frac{1}{2}n) - n$ leading to $(\frac{1}{2}n^2 - \frac{1}{2}n)$ without error AG	1	
7(f)	$m = 9$ cao	3	M1 for $\frac{1}{2}m^2 + \frac{1}{2}m = 5m$ A1 for $m^2 - 9m = 0$ or $m^2 = 9m$ or $m - 9 = 0$ or $m + 1 = 10$ or B2 for $[m = 9]$ $5m = 45$ and crosses = 45 or B1 for values for $5m$ and the number of crosses seen for at least $m = 7$ and 8 After 0, SC1 for answer 11

5. S18/qp21/q5

5(a)(i)	$6n - 5$ oe	2	M1 for $6n + k$ oe with $k \neq 0$
5(a)(ii)	256 is not exactly divisible by 6 or 247 in sequence and next one is 253 oe	1	
5(b)(i)	$p^2 - 3$ oe	1	
5(b)(ii)	$p^2 + 2p + 4$ oe	1	

6. W18/qp21/q8

8(a)	$n + 5$ $n + 10$	1	Both correct
8(b)(i)	$(n + 5)^2$ and $n(n + 10)$	M1	
8(b)(ii)	$n^2 + 5n + 5n + 25 - n^2 - 10n = 25$	A1	or $n^2 + 10n - n^2 - 5n - 5n - 25 = -25$
8(c)	63	3	M1 for $n + n + 5 + n + 10 = 174$ oe A1 for $n = 53$ If 0 scored, SC1 for answer 53